

Modeling the Detailed Conformational Effects of the Lactosylation of Hyaluronic Acid

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Cite This: *Biomacromolecules* 2025, 26, 541–555



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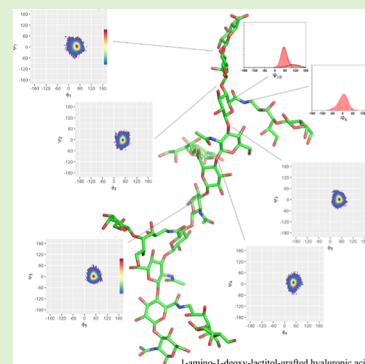


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ABSTRACT: Hyaluronic acid (HA) is a natural and biocompatible polysaccharide that is able to interact with CD44 receptors to regulate inflammation, fibrosis, and tissue reconstruction. It is a suitable chemical scaffold for drug delivery that can be functionalized with pharmacophores and/or vectorizable groups. The derivatization of HA is achieved to varying extents by reacting 1-amino-1-deoxy-lactitol via the carboxyl group to form amide linkages, giving rise to the grafted polymer, HYLACH. This retains the broad properties of HA, even though, as in most HA-grafted polymers, the detailed conformational effects of such substitutions, while crucial in the design or optimization of drug delivery systems, remain unknown. Here, the conformation, size, secondary structure, hydrogen bond network, and hydration features of lactosylated HA derivatives were evaluated by using multiple independent molecular dynamics simulations. This revealed subtle but nevertheless significant changes in the HA scaffold, establishing the density of grafting as the key parameter determining its properties.



1. INTRODUCTION

Hyaluronic acid (HA) is a linear, nonsulfated, polysaccharide made of repeating disaccharide units of (1–3) β -D-GlcNAc (1–4) β -D-GlcA (Scheme 1A). Hyaluronic acid or hyaluronate, as it should more properly be called since under physiological conditions the carboxyl group is in the anionic form, is an important constituent of the mammalian extracellular matrix (ECM) and connective tissues.^{1,2} The economic interest in HA is continuously growing and is driven largely by the cosmetics and skincare industries in which HA serves roles as “skin fillers” to combat the effects of aging and whose action is thought to combine both a physical filling process with biochemical antioxidant effects.^{3,4} HA also plays a significant role in tissue hydration and elasticity owing to its capacity to form viscous solutions, retaining significant quantities of water, and the resulting viscoelastic properties. In addition, HA is anti-inflammatory and antiangiogenic and is involved in cellular activities that include proliferation, migration, and adhesion⁵ through the interaction with several HA receptors, including the cell surface receptor cluster of differentiation-44 (CD44).^{6–10} Owing to its physicochemical properties, biodegradability, and biocompatibility, HA is used widely in pharmaceutical applications.^{11,12} It is particularly suitable as a drug delivery system because of its specificity toward several receptors such as CD44 and RHAMM (receptor for hyaluronic mediated motility),^{13,14} whose internalization mechanisms and the enzymatic degradation of the carrier HA contribute to intracellular delivery.¹⁵ CD44 receptors are overexpressed in tumor cells, suggesting potential

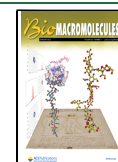
HA roles in the targeted delivery of anticancer drugs. The chemical structure of HA allows facile functionalization and modification, through both carboxyl (to form esters or amides) and hydroxyl groups,¹⁵ while amine groups can also be derivatized following removal of *N*-acetyl moieties. A broad range of HA-based polymers were synthesized by chemical methods in the form of conjugated or cross-linked HA for the enhancement and control of its therapeutic action.¹¹ HA grafted with poly lactic acid exhibits higher lipophilicity, providing greater affinity toward hydrophobic substances, corresponding to increased viscosity¹⁶ and decreased water solubility. Pharmacophore polymers, such as derivatives of HA in the form of HA–bilirubin nanomedicines, have been proposed for the modulation of dysbiosis during colitis,¹⁷ while a fluorescent form of HA intended for cell labeling, also stimulates cells.¹⁸ Pharmacophore polymers have been prepared conjugating the natural antioxidant: gallic acid, caffeic acid, ferulic acid, and the antimicrobial peptide *nisin* to HA.¹⁹ Polyphenols act as radical scavengers, while *nisin* was active against bacteria such as *S. epidermidis*, *S. aureus*, and *P. aeruginosa*,²⁰ and bacterial biofilms following covalent immobilization to a surface.^{21,22} HA, grafted with poly(*N*-

Received: September 26, 2024

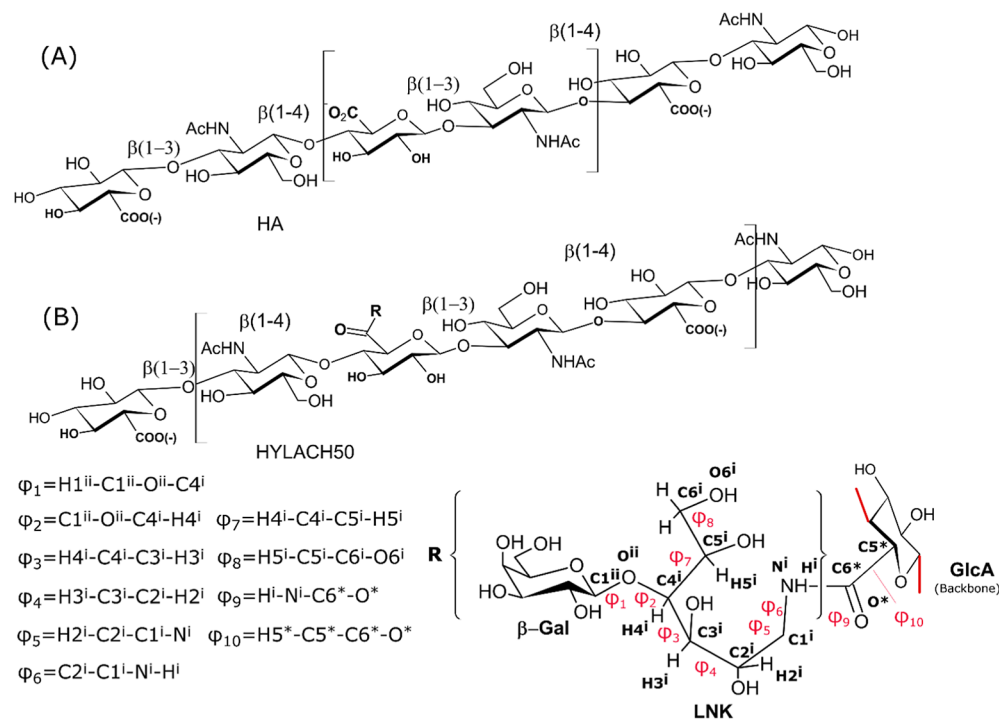
Revised: December 4, 2024

Accepted: December 5, 2024

Published: December 16, 2024



Scheme 1. (A) Scheme of the Connectivity of Hyaluronic Acid



The repeating unit is defined between square brackets $[-\text{GlcA-}\beta(1\text{-}4)\text{-GlcNAc-}\beta(1\text{-}3)]_n$. (B) Scheme of the connectivity of the grafted polymer HYLACH with degree of substitution (DS) = 50% (HYLACH50). The grafted group LACH-NH₂ (labeled “R”) is reported in curly brackets with the set of dihedral angles φ_1 to φ_{10} underlined in red, characterizing its conformations. The definition of these dihedral angles is specified in the bottom left corner. The linker LNK subunit connecting β -Gal with GlcA is underlined. The red segments indicate the C1 and C4 atoms of GlcA corresponding to the backbone of HYLACH.

vinyl imidazole), has been synthesized, characterized, and successfully tested against hepatic and breast cancers.²³ The imidazole moiety presents a broad range of activities, including antimicrobial, antifungal, and anticancer²⁴ but also gene regulation activity, promoting alkylation of DNAs.²⁵ HA oligosaccharides grafted with dihydrazides allowed the successful functionalization with ibuprofen and hydrocortisone drugs,²⁶ while a different strategy has involved the application of amino acid linkers to conjugate drugs such as paclitaxel with a HA scaffold.²⁷

Despite this wide and heterogeneous range of structures and applications, their size, conformation, average position, and orientation of the linked active pharmacophores groups, and how these affect the HA backbone, are crucial but still poorly investigated aspects of these grafted HA copolymers.

Previously, we reported the preparation and characterization of HA derivatives by functionalization of the carboxylic groups with 1-amino-1-deoxy-lactitol (LAC-NH₂)²⁸ to generate a product with the commercial name HYLACH, designed for the localization of galectin-3 (Gal-3). Galectins are soluble proteins based on conserved β -galactoside-binding sites located within their carbohydrate recognition domains.²⁹ Galectins are involved in a plethora of physiological events controlling intra- and extracellular processes, including growth, proliferation, and differentiation, thanks to their capacity to cross-link glycoproteins at the cell surface and in intracellular compartments.^{30–32} Galectins, and Gal-3, are involved in inflammatory diseases^{33,34} and participate in several pathophysiological processes, including fibrosis.^{35–37} Also, Gal-3 is overexpressed in most types of cancers; therefore, inhibition of

Gal-3 is increasingly considered as a promising broad therapeutic approach.³⁸

The scope of this study is the structural and conformational characterization of HA derivatives (HYLACH), in which the carboxyl group is functionalized with LAC-NH₂, and how these molecular features depend on the DS. Structure and conformational properties of HA have already been investigated using complementary approaches and varying conditions: X-ray diffraction of stretched fibers,³⁹ NMR and MD simulations in an aqueous solution,^{40–42} and by alternative methods for conformational exploration.^{43,44} The intramolecular hydrogen bond network between adjacent residues was found to contribute to the conformational stiffness of HA, its effective size, and its secondary structure.⁴¹

Three decasaccharides of HA and grafted HA, the latter characterized by a DS of 50 and 100%, and labeled HA, HYLACH50 and HYLACH100, respectively, were built. The *state-of-the-art* GLYCAM06 force field for carbohydrates was used in combination with the TIP3P water model; the 1-amino-1-deoxy-lactitol-grafted GlcA residue was previously parameterized according to a preliminary conformational analysis to determine its preferred conformation and the corresponding parameters according to the GLYCAM06 scheme. The conformational properties of the backbone and side chains were explored by comparing HA, HYLACH50, and HYLACH100 using MD simulation spanning time windows between 385 and 971 ns. To improve the efficiency of the *phase space* sampling of these systems, each molecule was submitted to two independent MD simulations setting different initial conformations of the backbone. The conformation of the backbone of these oligosaccharides, that of

the grafted groups, their intramolecular hydrogen bond networks, and their hydration features were investigated. Furthermore, the size and helical parameters were calculated and correlated with DS, obtaining a useful structural and conformational analysis of a class of derivatized HA, which, to the best of our knowledge, has not been analyzed at this level. We believe that these structural and conformational analyses, focused on grafted HA, are crucial both for improved scientific understanding and for the development and tailoring of polymeric drug delivery systems.

2. MATERIALS AND METHODS

2.1. HA and HYLACH Oligosaccharide Model Setup and Parameters of MD Simulations. Three decasaccharides representing the polymers HA and HYLACH at a DS of 50% and 100%, here labeled HA, HYLACH50 and HYLACH100 (Scheme 1), respectively, were created setting the initial conformation of GlcNAc and GlcA as chair 4C_1 , as already known in HA polymers in the solution state^{45–48} and in stretched fiber.³⁹ In these oligosaccharides, the residues are numbered from the nonreducing end as follows: GlcA(1), GlcNAc(2), GlcA(3), GlcNAc(4), GlcA(5), GlcNAc(6), GlcA(7), GlcNAc(8), GlcA(9), and GlcNAc(10). HYLACH50 is characterized by two grafted groups (LAC-NH₂) that are covalently linked through the amide bond to GlcA(3) and GlcA(7), while HYLACH100 has four grafted groups linked to GlcA(3), GlcA(5), GlcA(7), and GlcA(9). The conformation of the interglycosidic linkage $\beta(1-3)$ or $\beta(1-4)$ is defined by the pair of dihedrals: $\phi_i = H1_i-C1_i-OX_{i+1}-CX_{i+1}$, $\psi_i = C1_i-OX_i-CX_{i+1}-HX_{i+1}$, where $X = 3$ or 4 , respectively, while integer index $i = 1, 9$ indicates monosaccharide units from the nonreducing ($i = 1$) to the reducing end ($i = 9$). To improve the efficiency of the space of phase sampling of these systems, two independent MD simulations (replicas), characterized by different initial backbone conformation, here labeled CFG A and CFG B, were run for each of the HA, HYLACH50 and HYLACH100 oligosaccharides. In CFG A, ϕ_i/ψ_i was set to $54/-2$ and $37/-16^\circ$ in GlcA- $\beta(1-3)$ -GlcNAc- and GlcNAc- $\beta(1-4)$ -GlcA linkages, respectively; this was done in accord with the model of HA (1hya) in the PDB.³⁹ The alternative initial conformation CFG B is generated from the previous upon arbitrary perturbation of each glycosidic dihedral by 10° and therefore characterized by ϕ_i/ψ_i of $44/-12$ and $27/-26^\circ$ in the interglycosidic junctions GlcA- $\beta(1-3)$ -GlcNAc- and GlcNAc- $\beta(1-4)$ -GlcA, respectively. The initial conformation of the *N*-acetyl groups was set *trans*; in fact, this angle was reported to be between -150 and -160° in all the five GlcNAc of HA oligosaccharides, according to refs 39, 41, 42, 46, and 49. The initial conformation of the grafted groups LAC-NH₂ was set according to the conformation “A” reported in Figures S1 and S2 and Table S1, as determined separately upon conformational analysis (see the section **Parameterization of the Grafted GlcA in the Supporting Information**). Therefore, the initial conformation of each grafted group was set according to the following dihedrals: $\phi_1 = 60^\circ$, $\phi_2 = -64^\circ$, $\phi_3 = 73^\circ$, $\phi_4 = 76^\circ$, $\phi_5 = 63^\circ$, $\phi_6 = 0^\circ$, $\phi_7 = -175^\circ$, $\phi_8 = -63^\circ$, $\phi_9 = 180^\circ$, and $\phi_{10} = 111^\circ$ (Scheme 1). The conformation of the β -Gal residue has been set as chair 4C_1 ,⁵⁰ as verified by NMR spectroscopy in a comparable grafted polymer.⁵¹ The simulation cell box for each oligosaccharide was obtained surrounding the solute macromolecule by a 15 Å wide layer of water molecules. No counterions are included in these oligosaccharide models since nonbonded forces (electrostatic and dispersive) were described using the cutoff (12 Å) approach. The parameters and topology files for HA and HYLACH decasaccharides have been generated using AmberTools18,⁵² while the state-of-the-art force field for carbohydrate GLYCAM06 and the TIP3P⁵³ water model were applied. The GLYCAM06⁵⁰ parameters describing the grafted GlcA residue were set up following the procedure in the section **Parameterization of the Grafted GlcA in the Supporting Information**. The MD simulation application NAMD 2.14⁵⁴ and the VMD 1.9.3 visualization software⁵⁵ have been used to run MD simulations and for the analysis of the

sampled trajectories, respectively. The six simulation boxes, including HA, HYLACH50, and HYLACH100 decasaccharides in both initial conformations CFG A and CFG B, were independently submitted to the following simulation steps. The *first step* is energy minimization using 2×10^5 steps of the default energy minimization algorithm in NAMD 2.14. The *second step* corresponds to MD simulation in NPT conditions (Constant Number of particles, Pressure, and Temperature) to adjust the density of the simulation box, whose length was between 10 and 30 ns, according to the time required to reach a constant average volume of the cell. The temperature (300 K) and pressure (1.01325 bar) were controlled by a Lowe Andersen thermostat and Nosé–Hoover–Langevin piston algorithms, respectively. In this cell density equilibration phase, the initial conformation of the oligosaccharides was preserved by the harmonic potential energy restraint term (a harmonic constant of $0.5 \text{ kcal mol}^{-1}$) that has been applied to all atoms of the solute macromolecule until the cell volume has been shown convergence. After the cell density equilibration stage, the harmonic potential energy restraint was turned off, and the cell box of HA, HYLACH50, and HYLACH100, each of them in CFG A and CFG B initial conformation, were submitted to unconstrained MD simulation (*third step*) of length 385 and 415, 571 and 580, and 689 and 970 ns, respectively. In this *third step*, the conformation of each monosaccharide is followed plotting vs time the dihedral angle $H1-C1-C2-H2$ (Figure S3a–f). In fact, values of this dihedral proximal to $\pm 180^\circ$ (*trans*) indicate a 4C_1 chair conformation of the pyranose ring, while values proximal to 145° indicate a distorted boat conformation. Furthermore, the conformational equilibration period of HA HYLACH50 and HYLACH100 has been estimated analyzing the time evolution of the whole set of interglycosidic dihedrals, $\phi_i(t)/\psi_i(t)$ ($i = 1, 9$), and the *end-to-end* distance; in fact, these correspond to a reduced set of degrees of freedom that summarize the conformational behavior of almost linear oligosaccharide at short- and long-time scales. The *end-to-end* distance, which is measured between the C4 of GlcA(1) and C1 of GlcNAc(9),⁵⁶ corresponds to the longest size of these polymers and one of the slowest evolving degree of freedom. Collectively, the *end-to-end* distance and the interglycosidic dihedrals present a drift from their initial to the final “equilibrium” condition of approximate length 150, 200, and 400 ns for the simulated HA, HYLACH50, HYLACH100, respectively, and considering CFG A and CFG B replica, in which each system has been simulated (Figure S3g–i). Therefore, these time intervals were used as the conformational equilibration time of HA, HYLACH50, and HYLACH100 simulation boxes, respectively, while the remaining time is the *production* MD simulation. The equilibration and production MD simulation trajectory has been sampled every 10 ps. The conformational and statistical analyses of HA, HYLACH50, and HYLACH100 were applied on the production MD simulation, upon merging the trajectories generated from the two independent replica CFG A and CFG B of each system.

2.2. Backbone Conformation Analysis of HA and HYLACH Oligosaccharides. The conformation of HA, HYLACH50, and HYLACH100 is summarized by the set of interglycosidic dihedral angles ϕ_i/ψ_i . These dihedrals that were sampled in the production MD simulation were plotted as Ramachandran plots, correlating angle ϕ_i with the adjacent ψ_i . A density color map, whose gradients from blues to reds indicate states ϕ_i/ψ_i of increasing population density, has been plotted above the Ramachandran plots, allowing the determination of the most populated states. The statistical software R was used for this analysis.⁵⁷

2.3. Grafted Group Conformation Analysis. The 1D binning procedure (Origin 8.0) was applied to estimate the most populated states for the dihedral angles of *N*-acetyl groups of glucosamines and for the dihedrals of the grafted groups (LAC-NH₂) ϕ_i in HYLACH50 and HYLACH100. In this case, histograms with a bin size of 2° were used. When for the dihedral ϕ_i more than one state is visited, the relative population has been calculated using the *Integrate multiple peak routine* (Origin 8.0). In selected cases, in which the distribution of ϕ_i is represented by two or more superposed peaks, the enveloped distributions were fitted by using a superposition of two or more Gaussian functions using the *multiple peak fit routine* (Origin 8.0).

Table 1. Most Populated States of the Interglycosidic Dihedrals ϕ_i/ψ_i ($i = 1, 9$) of HA, HYLACH50, and HYLACH100 Oligosaccharides, as Predicted by the Ramachandran Plots and Density Color Maps^a

	ϕ_1/ψ_1	ϕ_2/ψ_2	ϕ_3/ψ_3	ϕ_4/ψ_4	ϕ_5/ψ_5	ϕ_6/ψ_6	ϕ_7/ψ_7	ϕ_8/ψ_8	ϕ_9/ψ_9
linkage	1→3	1→4	1→3	1→4	1→3	1→4	1→3	1→4	1→3
HA	45/−3	47/8	47/−7	45/15	45/−4	45/−6	43/−6	43/−10	43/3
HYLACH50	43/4	45/−3	45/−4	47/4	43/−10	50/−3	45/−4	47/2	45/−3
HYLACH100	43/7	45/−4	45/5	47/8	47/2	47/−6	45/1	47/1	52/−2
2BVK		48/7	50/9	48/7	50/10	48/7	50/10	48/7	
1HYA		37/−16	54/−2	37/−16	54/−2	37/−16			

^aAngles are in degrees. 2BVK and 1HYA are models of HA predicted in an aqueous solution⁴² and in stretched fibers.³⁹

The quality of the multiple peak fit is evaluated using the corresponding statistical parameter R^2 , which was found between 0.9881 and 0.9993.

2.4. Intramolecular Hydrogen Bond Analysis. The hydrogen bond (h-bond) analysis of HA and the grafted polymers was done focusing on their backbone, as proposed by Almond et al.,⁴² and further extended to consider potential h-bonds between the grafted groups and the backbone, and between different grafted groups. These h-bonds were characterized by their population percentage in the production stage of the MD simulation for HA, HYLACH50, and HYLACH100. Upon visual inspection of the MD simulation trajectories, the proton donor and acceptor groups that are potentially involved in h-bonds X-H...Y were selected, and a diagram correlating the distance between the proton donor and acceptor groups $d(X-Y)$ vs the angle between them $a(X-H-Y)$ was drawn. The cutoff conditions on the distance $d(X-Y) \leq 3.0$ Å and angle $150^\circ \leq a(X-H-Y) \leq 180^\circ$ define a restricted area on the lower right part of the diagram, corresponding to the existence condition of a *strong* h-bond, according to Almond et al.⁴² The ratio between the number of points inside this area and the total number of states gives the population percentage of the h-bond. This analysis and plots were performed using Origin 8.0 software. The average residence time of the selected intramolecular h-bond has been estimated counting the average number of consecutive snapshots in which the selected bond is found and multiplying this number by the time interval between two consecutive frames (10 ps). The error in the estimated residence time is not less than ± 5 ps.

Intramolecular h-bonds that are mediated by water molecules (water bridges) are detected using the *cpttraj* application that is included in the Ambertools package⁵² (versions 2024) and upon setting their geometrical condition on distance and angle that were defined before (strong h-bond).

2.5. Secondary Structure Analysis of HA and HYLACH Oligosaccharides. The molecular size and secondary structure of HA, HYLACH50, and HYLACH100 were investigated by calculating the *end-to-end* distance and the parameters of the helical geometry for each MD simulation frame. The *end-to-end* distance has been plotted versus time, while the corresponding average values calculated on the production stage, and considering both CFG A and CFG B replicas, have been reported. In this case, the reported error on the mean is found smaller than the last decimal digit. The secondary structure analysis of these HA-based polymers has been done defining for each molecule, a statistical ideal chain formed by 10 consecutive segment vectors $R_1, R_2, \dots, R_{10}$ that connect the opposite glycosidic oxygens O4 and O1 delimiting each glucuronic and glucosamine residue according to Almond et al.⁴¹ and Sugeta et al.⁵⁸ Therefore, nine planar angles between two consecutive vectors R_i and R_{i+1} ($i = 1, 9$) and eight dihedral angles between three consecutive vectors R_i, R_{i+1} , and R_{i+2} ($i = 1, 8$) define (upon averaging) an ideal helix, in which two different segments' length r_1 and r_2 can be distinguished, whose relative orientation is defined by two angles γ_1 and γ_2 and by two dihedral angles τ_1 and τ_2 , as reported in the scheme of Figure 10 in Almond et al.⁴¹ The geometry of this ideal helix is summarized by the parameters θ, h , and n that were calculated using the equation and according to Almond et al.⁴¹ In that case, θ and h have the geometrical meaning of the axial *turn* and axial *rise* per disaccharide, respectively, while n corresponds to the number of residues per turn (helical fold).

2.6. Hydration Properties of HA and HYLACH Oligosaccharides Investigated by Radial Distribution Function $rdf(r)$.

The hydration properties of HA and HYLACH have been investigated by calculating the radial distribution function $rdf(r)$ ⁵⁹ of the water molecules in the surrounding of selected groups and/or atoms, as implemented in VMD 1.9.3 software⁵⁵ (eq 1).

$$rdf(r) = \lim_{dr \rightarrow 0} \left[\frac{V}{4\pi N_{\text{pair}}} \frac{p(r)}{r^2 dr} \right] \quad (1)$$

The variable r represents the distance between the selected atoms or group (reference) of the oligosaccharide and the oxygen atom of each H₂O molecule surrounding it, according to a distance cutoff (15 Å). N_{pair} denotes the number of potential atom pairs in the system, and V is the volume of the simulation cell. $rdf(r) dr$ reflects the concentration of water molecules in a spherical layer of infinitesimal thickness (between r and $r + dr$) centered on the reference atom. $rdf(r)$ is calculated for each oligosaccharide in the production MD simulations. The range of validity of $rdf(r)$ is between 0 and 8–10 Å, i.e., less than the nonbonded forces cutoff (12 Å).^{60,61}

3. RESULTS

3.1. Monosaccharides Conformational Characterization of HA, HYLACH50, and HYLACH100. All the GlcNAc and GlcA residues preserve their ⁴C₁ chair conformation, as previously observed in HA in a water solution^{42,49} and in the solid state.³⁹ The dihedral angles H1–C1–C2–H2 of each GlcA (1, 3, 5, 7, and 9) and GlcNAc (2, 4, 6, 8, and 10) residue are close to $\pm 180^\circ$, corresponding to the ⁴C₁ chair. Despite this, the nonreducing terminal GlcA(1) residues of HYLACH50 and HYLACH100 and the near-terminal GlcA(9) of HA were observed populating (as a minority) a distorted boat conformation, characterized by values of dihedral H1–C1–C2–H2 close to 145° . The β -Gal residues of all the grafted groups populate only the ⁴C₁ chair conformation in all the simulations, in agreement with LAC-NH₂-grafted chitosan⁵¹ (Figure S3a–f).

3.2. Conformational Characterization of the Backbone of HA, HYLACH50, and HYLACH100. As for the HA, the conformation of almost linear (grafted) decasaccharide HYLACH50 and HYLACH100 is summarized by the set of interglycosidic dihedral angles ϕ_i/ψ_i that have been plotted vs time in Figure S3g,h,i. A density color map, whose gradient from blue to red is proportional to the density of states, has been superimposed onto each Ramachandran plot, allowing the localization of the most populated dihedrals (Table 1 and Figure S4a,b). The conformation of the interglycosidic dihedrals of HA shows slight differences according to the linkage connectivity $\beta(1 \rightarrow 3)$ or $\beta(1 \rightarrow 4)$, and both linkages have ϕ_i/ψ_i values around $50/10^\circ$, as found by Almond et al.⁴² Interestingly, the linkages $\beta(1 \rightarrow 4)$ show greater flexibility than $\beta(1 \rightarrow 3)$, particularly at the ψ_i degree of freedom, suggesting that this half of the glycosidic linkages is less

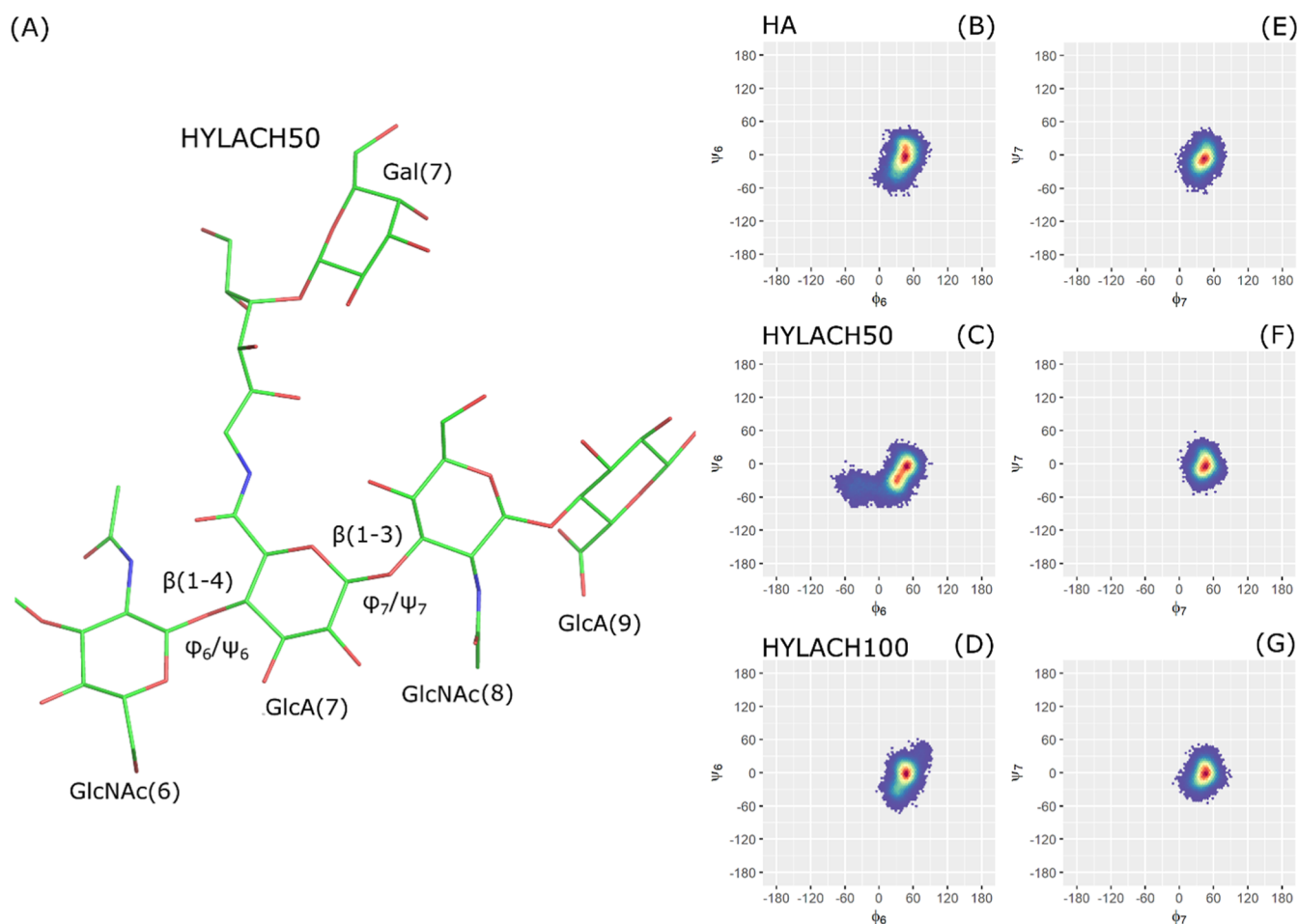


Figure 1. Local flexibility of the interglycosidic backbone of HA and the grafted polymer HYLACH at different DS. (A) Scheme of HYLACH50 showing LACH-NH₂ bound to GlcA(7) and surrounded by interglycosidic linkages ϕ_6/ψ_6 and ϕ_7/ψ_7 . The oligosaccharides are represented by sticks in which carbon, oxygen, and nitrogen atoms are colored green, red, and blue, respectively. (B–G) Ramachandran plot and density color map of the interglycosidic linkages ϕ_6/ψ_6 and ϕ_7/ψ_7 of HA (panels B, E), HYLACH50 (C, F), and HYLACH100 (D, G). The color density maps span between the most populated (reds) to the less (blues) populated states sampled.

restrained, again in accord with Almond et al.⁴² This correlates with slightly weaker hydrogen bonds in this half section of the $\beta(1 \rightarrow 4)$ interglycosidic linkage, as will be described later. The most populated ϕ_i/ψ_i states of HYLACH50 and HYLACH100 are almost comparable with those of HA, indicating that, despite the high DS (50% or 100%), the LAC-NH₂ moieties weakly affect the conformation of the backbone of these grafted oligosaccharides in comparison to HA (Table 1). Interestingly, HYLACH50 and to a lesser extent HYLACH100 also show $\beta(1 \rightarrow 4)$ linkages slightly more flexible than the neighboring $\beta(1 \rightarrow 3)$ as previously observed in HA (Figure S4a,b). The backbone conformational flexibility of HYLACH50 decreases significantly in proximity of the grafted GlcA(i); in fact, ϕ_i/ψ_i values show reduced flexibility in comparison to the adjacent linkages: ϕ_{i-1}/ψ_{i-1} and ϕ_{i+1}/ψ_{i+1} (Figures 1C,F and S4a,b). This local reduced flexibility of the backbone correlates with conformations in which the subunits LNK and Gal(i) are bent over the proximal GlcNac(i + 1), thereby restricting the ϕ_i/ψ_i degree of freedom in comparison to the neighboring ϕ_{i-1}/ψ_{i-1} and ϕ_{i+1}/ψ_{i+1} linkages (Figure 1A). This behavior is not observed in HYLACH100, in which all of the GlcA residues are grafted; therefore, all their ϕ_i/ψ_i degrees of freedom present the same local environment. Focusing on the $\beta(1 \rightarrow 3)$ linkage, the backbone flexibility of

these oligosaccharides increases slightly in the terminal (ϕ_1/ψ_1 and ϕ_9/ψ_9) in comparison to the internal sections (ϕ_3/ψ_3 , ϕ_5/ψ_5 , and ϕ_7/ψ_7), as can be observed comparing the size of the colored areas from yellow to red in Figure S4a,b. This effect, which is independent from the DS, is a general property of almost linear macromolecules and arises from the greater conformational entropy of the terminal section of the chains.^{62,63}

3.3. Conformational Characterization of the Grafted Groups in HYLACH50 and HYLACH100. In HYLACH50 and HYLACH100, the conformational properties of the grafted groups that include the β -Gal moiety, the linker LNK subunit, and an amide bond (Scheme 1B, are described by dihedrals ϕ_1 to ϕ_{10} , whose distributions were estimated and are reported as histograms (Figure S5a,b,c). From these distributions, the most populated ϕ_i values are reported in Table S3. When multiple states are allowed, their relative percentages were calculated as described (Materials and Methods).

Table S3 shows how the conformations of the grafted groups in HYLACH50 and HYLACH100 are similar, no matter what their position is on the backbone. Only slightly different population percentages are observed between grafted groups in the same molecule and between groups that belong to different

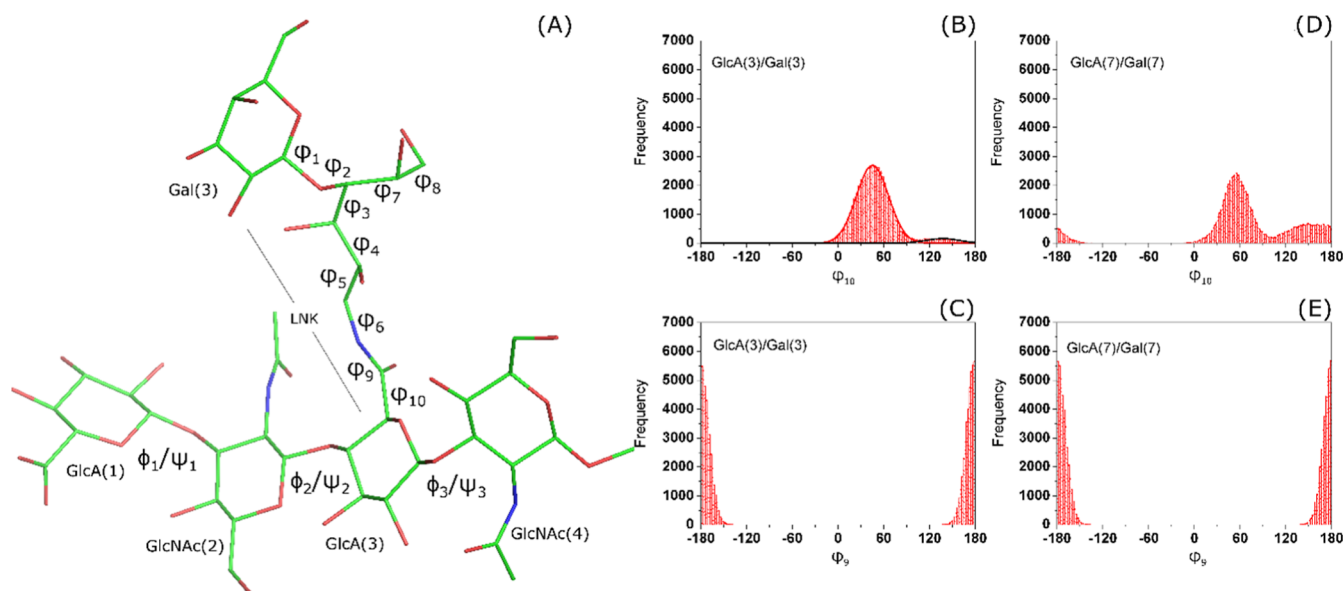


Figure 2. Conformation of the grafted groups in HYLACH oligosaccharides. (A) Set of dihedral angles ϕ_1 to ϕ_{10} that characterize the conformation of the linker (LNK) connecting GlcA(3) and GlcA(7) (backbone) to the corresponding β -Gal moiety in HYLACH50. Distribution functions of the dihedrals ϕ_{10} , ϕ_9 of GlcA(3) (panels B, C) and GlcA(7) (panels D and E) in HYLACH50 as sampled by MD simulation. The distribution of ϕ_{10} in panel (B) has been fitted by two Gaussian curves reported as red and black lines (Materials and Methods) to determine the relative percentage of the two states reported in Table S3.

molecules. The distributions of dihedrals ϕ_{10} and ϕ_9 of GlcA(3) and GlcA(7) in HYLACH50 are reported as histograms in Figure 2. The most significant variability in population percentages is shown by ϕ_{10} where the preferred *gauche+* conformation is between 94 and 69% in the two grafted groups in HYLACH50 and between 85 and 34% in the four grafted groups of HYLACH100. This suggests a pivotal role for ϕ_{10} in the motion exploring the conformation of LAC-NH₂-grafted groups of the HYLACH polymer. In HYLACH50, the dihedral ϕ_{10} of the grafted groups shows two possible states: 45° (94%), 137° (6%) in GlcA(3), and 55° (69%), 149° (31%) in GlcA(7). In both grafted groups, these two states correspond to a most favored *gauche+* and a less favored *trans* state, respectively. Almost an identical behavior is observed in the four grafted groups in HYLACH100, where ϕ_{10} populates preferentially the *gauche+* but with a smaller percentage than HYLACH50. This last observation suggests that the conformation of the grafted groups through ϕ_{10} is sensitively affected by DS.

In contrast, ϕ_9 assumes strictly $\pm 180^\circ$ corresponding to the most populated *trans* (or *anti*) state of a monosubstituted amide bond in all the simulated systems.^{64–67} Interestingly, ϕ_6 shows, among all the grafted groups (in both HYLACH50 and HYLACH100), a value close to 0°, corresponding to the *syn* conformation of secondary amides, which become dominant as the bulk of the *N*-substituent increases.⁶⁵ In this sequence, ϕ_7 and ϕ_8 describe the orientation of a side branch of the LNK subunit (Figure 2 A). The number of states populated by ϕ_8 increases slightly from that of HYLACH50 to that of HYLACH100 (Table S3).

In HYLACH oligosaccharides, following the distance between C4ⁱⁱ of β -Gal and C5* of the trivalent residue GlcA(*i*) allows the size of the grafted groups to be investigated.⁵¹ Interestingly, the grafted groups in both HYLACH50 and HYLACH100 populated almost *extended* conformations, characterized by distances between C4ⁱⁱ- β -Gal and C5*-GlcA(*i*) between 8 and 12 Å. In contrast, grafted

groups with *folded* conformations, in which these distances are smaller than 8 Å, are rarely sampled (Figure S6B,C). LAC-NH₂ groups in HYLACH100 populate preferentially *extended* conformations, even if the weight of *folded* conformations becomes slightly more significant, suggesting that the weight of *folded* vs *extended* conformations improves slightly with increasing DS (Figure S6 D).

3.4. Conformational Characterization of the Acetamido Side Chains in HA, HYLACH50, and HYLACH100.

The acetamido group of glucosamines in HA and HYLACH oligosaccharides is almost in *trans* conformation, with dihedral $\pm 180^\circ$, even if an additional state between -160 and -170° is also populated (Table S4), as underlined by a shoulder of the principal peak centered in the *trans* conformation $\pm 180^\circ$ (Figure S7).

3.5. Predicted Conformation of HA, HYLACH50, and HYLACH100.

Tables 1 and S3 summarize the most favored conformations of HA, HYLACH50, and HYLACH100, as estimated by MD simulations. The optimized conformation for each of these oligosaccharides (Figure 3A–C) was obtained by selecting a random structure of the oligosaccharides in the production stage of the MD simulation and setting all the dihedrals of the backbone (ϕ_i/ψ_i), those of the grafted groups (ϕ_1 – ϕ_{10}), and that of the acetamido groups with the corresponding most populated values reported in Tables 1, S3, and S4, respectively. Figure 3 shows that HA, HYLACH50, and HYLACH100 present weak differences in their backbone geometries; in fact, upon superposition, the RMSD between the backbone of HA and HYLACH50 is 2.59 Å and between HA and HYLACH100 is 2.89 Å. This suggests that lactosylation of HA weakly affects the backbone conformation of HA, and that this alteration grows progressively with DS, reflecting in a different helical geometry, as described later. Also, the *end-to-end* distance is significantly shorter in HYLACH100 than the two previous cases (Figure 3D, and paragraph 3.6). Figure 3A–C also reveals that the most populated states of dihedrals ϕ_1 to ϕ_{10} are compatible with

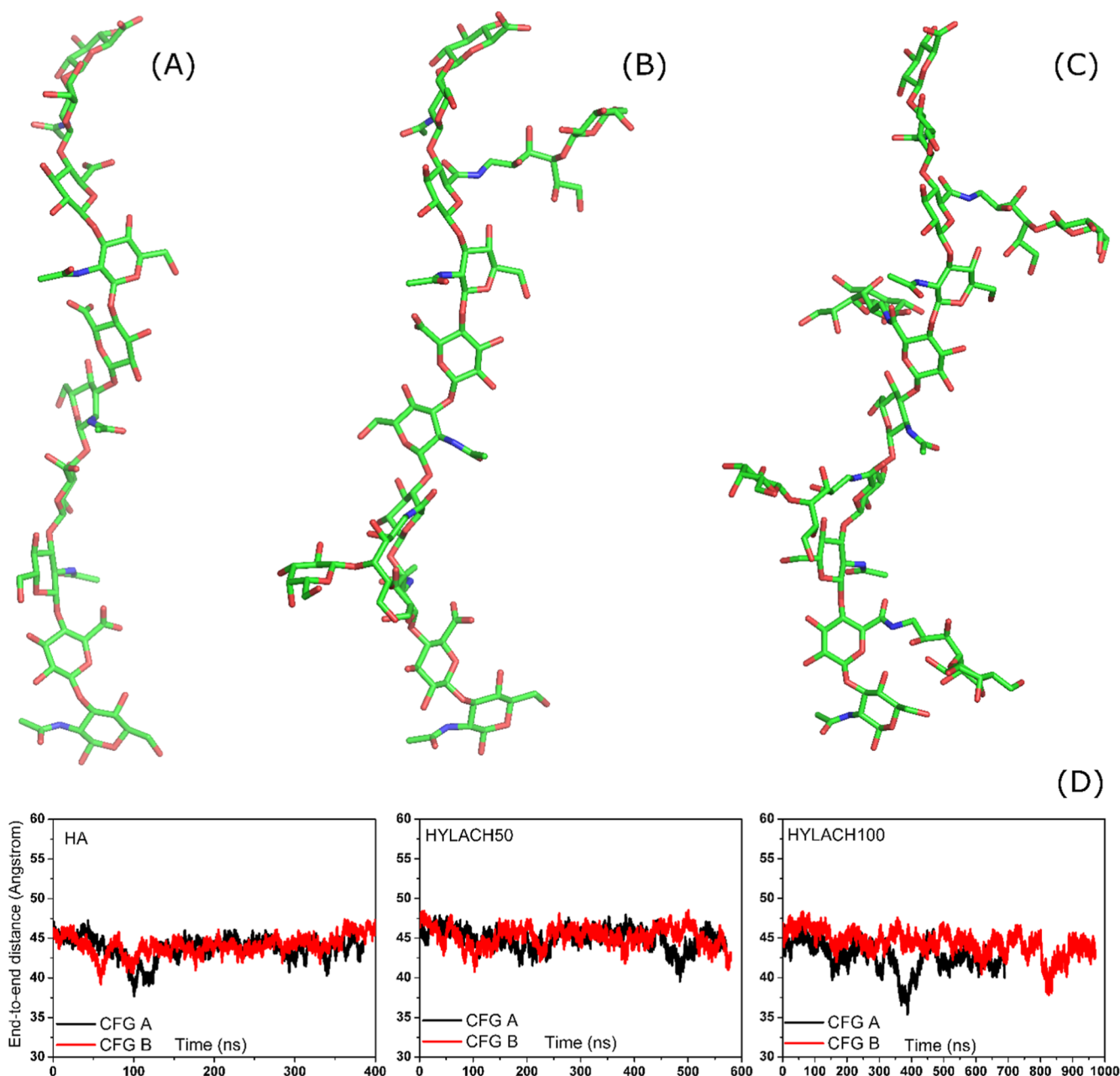


Figure 3. Predicted conformation and size of HA, HYLACH50, and HYLACH100 oligosaccharides. (A–C) Optimized conformation of the oligosaccharides HA, HYLACH50, and HYLACH100 setting all the known dihedrals of the backbone ϕ_i/ψ_i , that of the grafted groups $\varphi_1-\varphi_{10}$, and the acetyl groups, to the most populated values (Tables 1, S3, and S4, respectively). (D) *End-to-end* distance vs time of the oligosaccharides HA, HYLACH50, and HYLACH100 as sampled by two independent MD simulations starting from two different backbone conformations CFG A (black lines) and CFG B (red lines) of the oligosaccharides.

extended conformations of LAC-NH₂ groups in HYLACH50 and HYLACH100, in agreement with the analysis reported in Figure S6.

3.6. Molecular Size and Secondary Structure of HA, HYLACH50, and HYLACH100. The *end-to-end* distance quantifies the longest dimension of these almost linear oligosaccharides. By plotting the *end-to-end* distance vs time (ns) of HA, HYLACH50, and HYLACH100, the effects of grafted groups on the molecular size can be analyzed (Figure 3 D). The *end-to-end* distance does not change significantly in going from HA to HYLACH50, while it decreases slightly when comparing HA (or HYLACH50) with HYLACH100. In fact, the average value of the *end-to-end* distance is 44.3, 45.0,

and 43.1 Å in HA, HYLACH50, and HYLACH100, respectively. This suggests a modest attractive interaction between the grafted groups and the backbone, and/or between adjacent grafted groups, but only at sufficiently high DS. In fact, considering the flexibility that these grafted groups show in the spanned MD simulation times, the chance that these Gal or LNK subunits must come into contact with the backbone and/or with a proximal grafted group is likely to increase with DS, as will be described later.

The secondary structure has been further analyzed calculating the *axial turn* per disaccharide (θ), the *axial rise* per disaccharide (h), and the *number of residues* per turn (n), in accordance with an earlier work.^{41,58} These parameters were

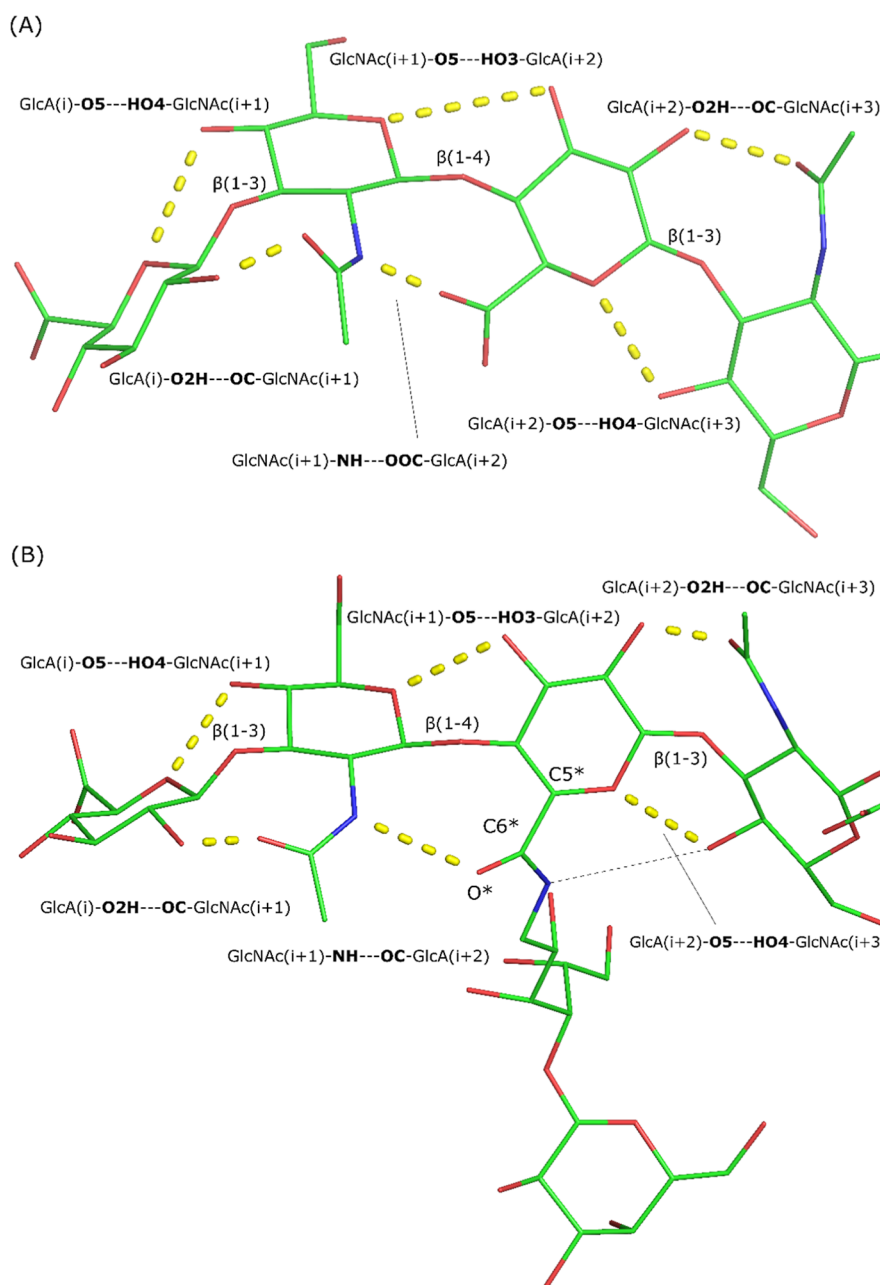


Figure 4. Most frequent hydrogen bonds (yellow dashed lines) in MD simulation of HA, HYLACH50, and HYLACH100 deca-saccharides that contribute to the polymer backbone conformation. The atoms that are involved as proton “donors” or “acceptors” are underlined in bold. The oligosaccharides are represented by sticks (carbon, oxygen, and nitrogen atoms are colored green, red and blue, respectively). Hydrogen atoms are not shown. (A) Scheme of the hydrogen bond network between GlcNAc(2), GlcA(3), and GlcNAc(4) of HA (B). Scheme of the h-bond network between GlcNAc(2), GlcA(3), and GlcNAc(4) of HYLACH50 and HYLACH100. The black dashed line indicates a potential h-bond or polar contact between the amide group = N^i-H^i and the facing hydroxyl HO4 of GlcA and the proximal GlcNAc (see text).

calculated by describing the sequence of GlcA and GlcNAc residues with a sequence of ideal vectors connecting oxygen O4 with O1 and oxygen O3 with O1, respectively (Materials and Methods). The secondary structure of HA is a left-handed helix characterized by values of $\theta = 116^\circ$, $h = 9.7 \text{ \AA}$, and $n = 3.1$, which is close to the 4-fold helix of HA oligosaccharides in water described by Almond et al.,⁴² in which the model PDB ID = 2BVK presents $\theta = 89^\circ$, $h = 8.3 \text{ \AA}$, and $n = 4.0$. The corresponding parameters in HYLACH50 are $\theta = 122^\circ$, $h = 9.8 \text{ \AA}$, and $n = 3.0$, indicating that the secondary structure of HA is almost unaffected by the LAC-NH₂-grafted groups at a DS around 50%. In contrast, HYLACH100 is characterized by

helical parameters: $\theta = 104^\circ$, $h = 9.5 \text{ \AA}$, and $n = 3.5$, suggesting sensitive contraction of the backbone helical pitch and a corresponding expansion of the helical section. Therefore, the geometry of the helix contracts in the axial direction; consequently, the number (n) of residues per turn increases, and the axial turn per disaccharide (θ) decreases, at high DS.

3.7. Hydrogen Bond Network of the Backbone of HA, HYLACH50, and HYLACH100. The potential h-bonds between adjacent monosaccharides of the backbone of HA, HYLACH50, and HYLACH100 were estimated by the application of the geometrical constraints that define a *strong* h-bond according to Almond et al.⁴² (Materials and Methods).

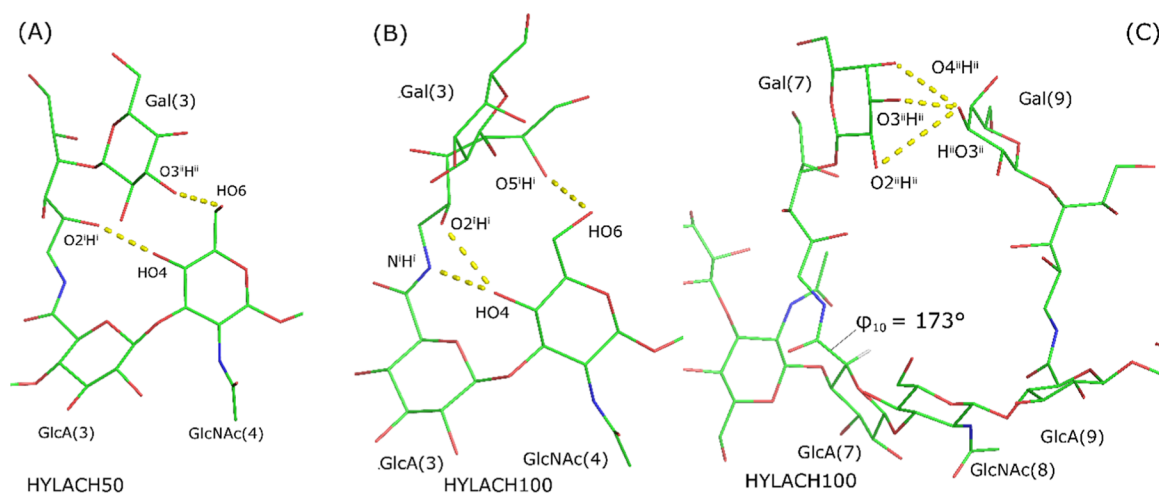


Figure 5. Potential hydrogen bonds between subunits of the grafted groups (LNK or β -Gal) and the backbone (A,B) or between two adjacent grafted groups (C). Snapshot of HYLACH50 (202.42 ns, panel A). HYLACH100 (500 ns, panel B) and HYLACH100 (550 ns panel C) extracted from the MD simulation showing potential h-bonds between the LNK and Gal(3) subunits of the grafted groups and the neighboring GlcNAc(4) (panels A and B). Potential h-bonds between two adjacent Gal(7) and Gal(9) subunits in HYLACH100 (panel C). The dihedral angle $\varphi_{10} = 173^\circ$ corresponding to a *trans* conformation, which allows Gal(7) and Gal(9) subunits to approach, is reported in panel C. The populations of these types of h-bonds (yellow dashed lines) and their average residence times are reported in Table S6.

As an example of the distance–angle correlation plots, which allows estimation of the h-bond population, we report the case of h-bond: GlcA(7)-O5---HO4-GlcNAc(8) in HA, HYLACH50, and HYLACH100 (Figure S8). In accord with Almond et al.,^{42,68} four different intramolecular h-bonds characterize the backbone of these oligosaccharides. Two hydrogen bonds span the $\beta(1-3)$ linkages: GlcA(*i*)-O5---HO4-GlcNAc(*i* + 1) and GlcA(*i*)-O2H---OC-GlcNAc(*i* + 1), and two span the $\beta(1-4)$ linkages: GlcNAc-O5(*i*)---HO3-GlcA(*i* + 1) and GlcNAc(*i* + 1)-NH---OOC-GlcA(*i* + 2), where the index *i* runs from 1 to 8,^{46,69} and the proton donor and acceptor groups are underlined in bold (Figure 4). Interestingly, the two h-bonds that involve the pyranoside oxygen and the facing hydroxyl group of the adjacent residues GlcA(*i*)-O5---HO4-GlcNAc(*i* + 1) and GlcNAc-O5(*i*)---HO3-GlcA(*i* + 1) show the highest population, the highest average residence times in all of the simulated oligosaccharides, and correspond to the strongest h-bonds. More precisely, the former hydrogen bond presents a slightly greater population in HA, HYLACH50, and (less evidently) HYLACH100 (Table S5), supporting greater flexibility of $\beta(1-4)$ compared to the $\beta(1-3)$ linkage, as a general property of these oligosaccharides, and as detected by Ramachandran plots in Section 3.2 (Figure S4A,B). By contrast, the two hydrogen bonds involve the acetamido groups of the glucosamines GlcA(*i*)-O2H---OC-GlcNAc(*i* + 1) and GlcNAc(*i* + 1)-NH---OOC-GlcA(*i* + 2), which are less populated and present shorter residence times; in fact, these acetamido groups are principally in *trans* conformation in solution and rarely meet the h-bond geometrical condition (Figure S7 and Table S4). For each simulated oligosaccharide, the populations of these interactions fluctuate locally along the backbone. In HA, the population of the h-bonds GlcA(*i*)-O5---HO4-GlcNAc(*i* + 1) and GlcNAc-O5(*i* + 1)---HO3-GlcA(*i* + 2) varies from 23–39% and 21–35%, respectively. Analogously, in HYLACH50, the population of these h-bonds ranges between 16 and 39% and 25–43%; in HYLACH100, the corresponding population ranges between 17 and 37% and 10–36%. By contrast, the weakest h-bonds GlcA(*i*)-O2H---OC-GlcNAc(*i* + 1) and GlcNAc(*i* + 1)-NH---

OOC-GlcA(*i* + 2) in HA show populations between 4 and 9%, in both cases. Furthermore, the same h-bond populations span 4–6% and 1–8% in HYLACH50, and between 2 and 5%, and 4–11% in HYLACH100 (Table S5). This indicates that the populations of the two strongest h-bonds that characterize the HA backbone are not affected or, in selected cases, are increased slightly by the grafted groups at low DS. In contrast, at high DS, the population of these h-bonds decreases. Differently, the population of the weakest hydrogen bonds is more sensitive to the grafting groups since their population decreases between DS 0 and 100%. A possible molecular explanation of these effect follows. Focusing on HYLACH50, the dihedral angle φ_{10} , which connects the LACH-NH₂ moiety to the backbone of HA in GlcA(3) and GlcA(7), collects the motion of the whole group and orients the polar amidic group = Nⁱ-Hⁱ toward the facing hydroxyl HO4- of GlcNAc(4) or GlcNAc(8), respectively, only when it assumes the *gauche*+ state. The *gauche*+ state of φ_{10} , which is favored in HYLACH50 (Table S3), promotes the polar contact between the amide group = Nⁱ-Hⁱ of GlcA(3) with the proximal hydroxyl HO4 of GlcNAc(4) (dotted line in Figure 4B), stabilizing the hydrogen bonds GlcA(3)-O5---HO4-GlcNAc(4) and indirectly GlcNAc(2)-O5---HO3-GlcA(3). Only very weak hydrogen bonds between the amide = Nⁱ-Hⁱ and the facing hydroxyl HO4 (population 0.1–0.2%) were observed in all of these grafted oligosaccharides (dotted line in Figure 4B and Figure S5B). Otherwise, the *trans* state of φ_{10} and the conformational transition *gauche*+ vs *trans* disfavor the geometrical conditions for these h-bonds. This supports an increase of the population of these h-bonds in going from HA to HYLACH50; by contrast, HYLACH100 that presents a further reduced *gauche*+ component of φ_{10} in average on its four grafted groups disfavors the population of these two types of h-bonds in comparison to HYLACH50 and HA. Finally, the same interpretation can be applied for the residence time of these long living (strongest h-bonds), whose average residence time increases in going from HA to HYLACH50 and decreases slightly from HYLACH50 to HYLACH100 (Table S5).

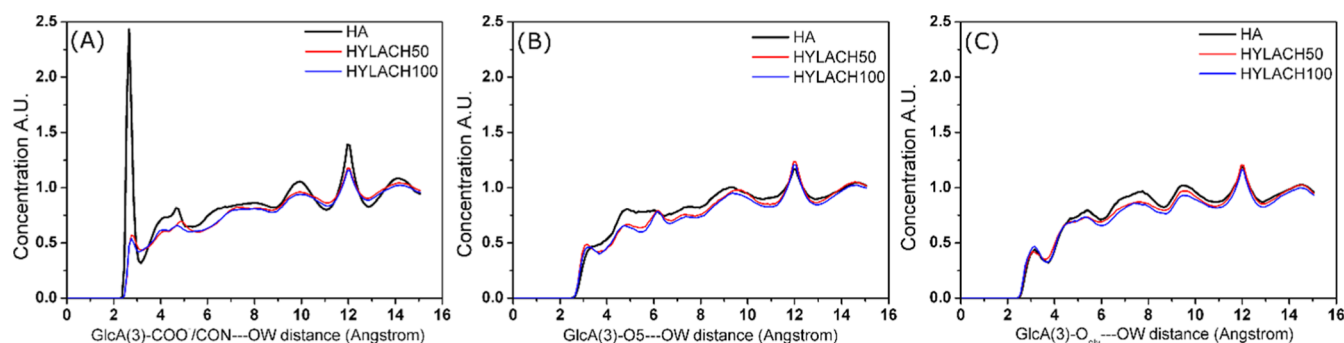


Figure 6. Radial distribution function $rdf(r)$ of water molecules surrounding selected groups of HA and HYLACH. (A) $rdf(r)$ of the distances between the oxygen atom of the carboxyl or oxygen and nitrogen of amide group of GlcA(3) and that of the water molecule. (B,C) $rdf(r)$ of the distance between intraring oxygen (O5) or the interglycosidic oxygen (O_{gly}) of GlcA(3) and the water molecules surrounding it, respectively. Black, red, and blue lines correspond to the $rdf(r)$ calculated for HA, HYLACH50, and HYLACH100, respectively.

3.8. Hydrogen Bonds of the Grafted Groups of HYLACH50 and HYLACH100. The same analysis has been extended to the h-bonds that the grafted groups, involving the linker (LNK) and/or the terminal subunit (β -Gal), potentially make with the residues of the backbone or the adjacent grafted groups in HYLACH oligosaccharides. Population percentages of these h-bonds greater than or equal to 1% are reported, with the corresponding average residence times in Table S6. As previously described, the LAC-NH₂ groups in HYLACH50 and HYLACH100 preferentially show *extended* rather than *folded* conformations; furthermore, the latter consistently present the LNK or the β -Gal subunits of the grafted GlcA(*i*) bent over its proximal GlcNAc(*i* + 1). Weak h-bonds between the hydroxyls of these subunits characterize their contact with the backbone of HYLACH50 and HYLACH100 (Table S6 and Figure 5A,B). Interestingly, the population of h-bonds between the β -Gal or the LNK moiety of GlcA(*i*) and the neighboring GlcNAc(*i* + 1) increases from HYLACH50 to HYLACH100, suggesting a significant role of DS in increasing the chance of contacts between the grafted groups and the backbone. No h-bonds were found between adjacent grafted groups in HYLACH50 since their accessible distances prevent any contacts between them. Only HYLACH100 shows hydrogen bonds between two adjacent grafted groups involving their multiple hydroxyls. In this case, the β -Gal moiety of GlcA(7) using some of its hydroxyls forms a transient h-bond network with that of β -Gal(9) (Table S6 and Figure 5C). These MD simulations predict that the contact between adjacent β -Gal(*i*) and β -Gal(*i* + 1) units is allowed only if the *pivotal* dihedral angle φ_{10} assumes its less populated *trans* state. Interestingly, the population of φ_{10} *trans* state, that is promoted by the van der Waals interaction between the hydrophobic group = N^HH'-C'H'2- of the LNK subunit connecting Gal(*i*) and GlcA(*i*), with the methyl group of the preceding GlcNAc(*i* - 1) (Figures 4B and S9B), increases sensitively as DS increases. Again, these observations suggest a sensitivity to DS in improving the frequency and, consequently, the strength of interactions between grafted groups.

MD simulation also predicts how the average residence time of the h-bonds between grafted groups and the backbone and between adjacent grafted groups is significantly greater than that of the backbone even though their population percentage is smaller than 10% (Table S6). This reflects the much slower dynamics of the grafted groups, whose periods are required to cover longer distances to engage or disengage contacts in comparison to that of the backbone.

3.9. Intramolecular Hydrogen Bonds That Are Mediated by Water molecules. HA and HYLACH oligosaccharides exhibit a detailed intramolecular h-bond network that is mediated by water molecules (water bridges) as previously reported.^{42,49} These are almost located between two adjacent residues and are presumed to reinforce the network of the intramolecular h-bond of the backbone HA and grafted HA oligosaccharides that has been analyzed in Section 3.7. Here, water bridges show population percentage up to 10% of the simulation time and are reported (when they exceed or are equal to 2%) in Table S7. The most heavily populated involve a water molecule H—O—H coordinating one of the carboxyl oxygens of GlcA(*i*) and O4 of GlcNAc(*i* + 1). Another frequent bridge localizes a water molecule between the amide NH of GlcNAc(*i*) and the faced carboxyl (or amide group in HYLACH) of GlcA(*i* + 1). Water bridges are also found between the C6—OH of GlcNAc and the proximal HO3 of GlcA. The network of water bridges is comparable in HA and HYLACH oligosaccharides, although lactosylation weakens some of them, as is particularly evident between GlcNAc(*i*)-NH and OOC-GlcA(*i* + 1), and between the consecutive pair GlcA(*i* + 1)-COO and O4-GlcNAc(*i* + 2), where the carboxyl is replaced by an amide group of the lactosyl moiety. The reduced ability of the amide group to coordinate water (see Section 3.10) and the partial hydrophobic character of the LNK are presumed to perturb the order of the proximal water molecules, inducing a local depletion of water. It is important to emphasize the way in which HYLACH oligosaccharides present water bridges connecting LNK and Gal subunits of the same grafted group but also that they connect the nearby GlcNAc. Interestingly, while the population of these h-bonds decreases with DS, their total number grows with DS (Table S7).

3.10. Hydration Properties of HA and HYLACH. The distribution of water molecules surrounding selected groups of HA and HYLACH was investigated using the radial distribution function $rdf(r)$. To understand how lactosylation affects the hydration of HA for selected atoms within these oligosaccharides, $rdf(r)$ was centered on the oxygens of the carboxylate (COO⁻) and the oxygen and nitrogen of the amide (CON) group; the latter replacing the former in HYLACH. We also focus our attention on the intraring oxygen (O5) atom of GlcA(*i*) and the interglycosidic oxygens. Figure 6 reports this analysis for residue GlcA(3), but comparable results were obtained for intrachain residues GlcA(5) and

GlcA(7), while slightly higher hydration properties were observed for the terminal GlcA(1) and penultimate GlcA(9).

The carboxyl of HA shows the highest concentration of the water in both the first (2.7 Å) and the wider second hydration shells (4.2–4.7 Å), whose position is in accord with the water distribution reported by Kauffmann et al.⁷⁰ Interestingly, the corresponding amide group in HYLACH50 and HYLACH100 presents a significantly reduced concentration of water in both of these hydration shells and at longer distances (Figure 6A). This describes the reduced efficacy of the amidic group in coordinating the water molecules compared to the carboxyl group. The intraring oxygen of GlcA in HA has the first and the wider second hydration shell located at 3.4 and 4.8 Å, respectively, but the amount of water that they can coordinate is smaller than that of carboxyl groups. Interestingly, the proximal amide group increases slightly the water concentration in this first shell, while, in the second shell and at longer distances (6–8 Å), the surrounding water is depleted in both HYLACH50 and HYLACH100 (Figure 6A). Analogously, the interglycosidic oxygen between GlcA(3) and GlcNAc(4) shows a depletion of water from the second shell (4.6–5.4 Å) to longer distances (6–8 Å) (Figure 6C). These results suggest that upon lactosylation of HA, the water concentration in the surrounding of the amide and all these proximal oxygens are significantly depleted at longer distances 4–8 Å; in this case, the hydrophobic character of the linker is presumed to repel water molecules in that range of distance.

4. DISCUSSION

Three decasaccharides HA, HYLACH50, and HYLACH100, characterized by different DS of 0%, 50%, or 100%, were simulated for time intervals from 385 to 971 ns. The chair ⁴C₁ conformation of GlcA, GlcNAc, and β-Gal was preserved in all these oligosaccharides even if the terminal GlcAs were observed to weakly populate a distorted boat conformation.

The interglycosidic dihedrals ϕ_i/ψ_i in HA were found to be in agreement with the corresponding most populated states of tetra-, hexa-, and octasaccharide of hyaluronic acid in a water solution, previously investigated by MD simulation and NMR spectroscopy^{42,45} and by restrained MD simulation,⁴⁶ while those conformations differed slightly from that of hyaluronic acid obtained by X-ray diffraction of oriented fibers.³⁹ Interestingly, grafted oligosaccharides HYLACH50 and HYLACH100 presented interglycosidic conformations that weakly differed from that of HA, despite DS changes from 0 to 100% (Table 1). Their β(1–4) interglycosidic linkages are slightly conformationally more flexible than the neighboring β(1–3) linkages, while this effect diminishes at high DS; this correlates with a slightly stronger h-bond O5---HO4 across the β(1–3) linkage, in comparison to the proximal O5---HO3 at the β(1–4). Furthermore, at intermediate DS, the flexibility of glycosidic linkages decreases locally in proximity to the grafted GlcA(*i*). In fact, the grafted groups of GlcA(*i*) (infrequently) assume *folded* conformations in which the β-Gal moiety bends over the proximal GlcNAc(*i* + 1), restricting the flexibility of ϕ_i/ψ_i as observed in HYLACH50 and less evidently in HYLACH100 (Figures 1 and S4a,b).^{62,63}

The conformations of the grafted groups are almost comparable in HYLACH50 and HYLACH100, as depicted by dihedrals from ϕ_1 to ϕ_{10} , while sensitive changes become evident in ϕ_{10} at high DS (Table S3). In fact, ϕ_{10} shows greater variability in all of these grafted groups and can be seen as *pivot* dihedrals connecting the more flexible grafted groups with the

more rigid backbone. In this case, ϕ_{10} samples two states *gauche+* and *trans*; the *gauche+* is always favored in these grafted polymers since it promotes the strong h-bonds of the backbone GlcA(*i*)-O5---HO4-GlcNAc(*i* + 1) and GlcNAc-O5(*i* + 1)---HO3-GlcA(*i* + 2) even though its preference decreases (slightly) at high DS. By contrast, the less favored *trans* state of ϕ_{10} allows contact between two adjacent grafted groups in grafted-HA at high DS and the van der Waals contacts between the LNK subunits and the methyl groups of the preceding GlcNAc (Table S3 and Figures 4 and S9B).

The secondary structures of HA and grafted HA, corresponding to the size and helical geometry of the backbone, were also investigated. The *end-to-end* distance of HA, HYLACH50, and HYLACH100 predicted by MD simulation is comparable at low DS,⁷¹ while becoming slightly smaller at high DS (Figure 3). This indicates that attractive interactions between the grafted groups and the backbone and/or between two adjacent grafted groups (using their multiple exposed hydroxyls) are prevalent on repulsive excluded volume effects. In fact, under good solvent conditions, these repulsive effects would expand the size and stiffness of highly density grafted macromolecules, which contrasts the observed behavior.^{62,72,73} LAC-NH₂ groups in these grafted polymers show predominantly *extended* rather than *folded* conformations, as was observed in the polymer CHITLACH, a chitosan grafted with 1-amino-1-deoxy-lactitol;⁵¹ also, these MD simulations reveal that the percentage of the *folded* conformations increases sensitively with DS (Figure S6). In the *extended* conformation, the β-Gal moiety is fully solvated and accessible from the outside of the macromolecule and corresponds to the condition required for the β-Gal moiety to be recognized by Gal-3. In the *folded* conformation, the subunit β-Gal and/or the LNK bends toward the backbone, allowing transient hydrogen bonds between grafted group subunits (LNK or β-Gal) and backbone (rarely reinforced by water bridges), which were found in HYLACH50 and, to a greater extent, in HYLACH100 (Figures 5 and S6 and Table S6), suggesting that this propensity of the grafted groups to form intramolecular interactions increases with DS.

The secondary structure of the HA decasaccharide in water is a left-handed 3-fold helix comparable with the 4-fold and slightly more contracted helix of tetra-, hexa-, and octasaccharide of hyaluronic acid.^{42,43} Comparable 3–4-fold helical secondary structures were found for hyaluronic acid in the solid state and with different counterions,^{39,69} while the most extended 2-fold helix was found for this polymer at low pH.^{43,74} At low DS, the grafted HA preserves the helical geometry of HA. At high DS, the attractive forces between the LNK and/or the β-Gal moieties with its proximal GlcNAc of the backbone contract the helix of the HYLACH slightly in an axial direction, widening its section compared to HA, while preserving its 3-fold helical character. Interestingly, this size contraction in term of *end-to-end distance* and helical geometry that decasaccharides of grafted HA exhibit as DS increases qualitatively correlates with the observed reduction of viscosity (η) and of hydrodynamic radius (R_h), which were observed on a larger scale polymer HYLACH (see Table 3 in ref 28).

The intramolecular hydrogen bond network of the backbone of HA and HYLACH involves four different h-bonds in a structural repeating unit that includes β(1–3), followed by β(1–4) linkage.⁴² Two hydrogen bonds characterize the β(1–3) linkage, GlcA(*i*)-O5---HO4-GlcNAc(*i* + 1) and GlcA(*i*-

$\text{O2H} \cdots \text{OC-GlcNAc}(i+1)$, and two characterize the $\beta(1\text{--}4)$ unit, $\text{GlcNAc-O5}(i+1) \cdots \text{HO3-GlcA}(i+2)$ and $\text{GlcNAc}(i+1)\text{-NH} \cdots \text{OOC-GlcA}(i+2)$ (Figure 4). Notably, the most populated and longest lasting of these h-bonds in HA, but also in grafted-HA, involve the pyranoside oxygen and the facing hydroxyl group of the adjacent residue; precisely, $\text{O5} \cdots \text{HO4}$ is also slightly stronger than its homologue $\text{O5} \cdots \text{HO3}$, supporting the slightly greater flexibility of this latter interglycosidic linkage. The h-bond $\text{O5} \cdots \text{HO3}$ across the $\beta(1\text{--}4)$ is comparable to that found by Braccini et al.⁷⁵ in glucuronan [$\beta\text{-D-(1--4)GlcA}$] oligosaccharides and in CHITLACH oligosaccharides of Esteban et al.⁷⁶ The h-bonds that involve the carbonyl or the NH moiety of each acetamido group of glucosamines present a significantly smaller population and shorter residence times; this correlates with the preferred *trans* conformation that these acetamido groups show in all these oligosaccharides and that was found in agreement with hyaluronic acid in water.^{41–43,49} Despite this, these contacts are mediated by water molecules that still contribute to the conformation of the backbone (Table S7). Overall, this analysis shows how the h-bond network of the backbone of LAC-NH_2 -grafted HA is comparable to that found in HA, even though at high DS, the grafted groups affect it sensitively.

MD simulation predicts that the frequency of transient hydrogen bonds between the grafted groups through the subunit LNK and/or $\beta\text{-Gal}$, with the adjacent GlcNAc of the backbone, increases with DS (Figure 5A,B and Table S6). MD simulation also shows that interactions between two consecutive 1-amino-1-deoxy-lactitol groups in grafted HA with DS lower than 50% are not allowed because the accessible distances between them prevent the formation of hydrogen bonds. In contrast, at high DS, weak hydrogen bonds between two consecutive $\beta\text{-Gal}$ moieties using their set of hydroxyls were observed (Figure 5C and Table S6), and, as previously described, require the pivotal angle φ_{10} to assume the less favored *trans* state, whose population increases sensitively with DS, thereby allowing the establishment of an alternative balance between the intermolecular forces of the backbone and those of the grafted groups. Therefore, DS can be interpreted as a crucial structural parameter modulating the size, the helical geometry, the intramolecular hydrogen bond network, but also the recognition propensity of the $\beta\text{-Gal}$ drugs, in LAC-NH_2 -grafted HA.

HYLACH remains a water-soluble polymer; however, in comparison to HA, it is predicted to have a significant depletion of water concentration in the proximity of the amide groups, which upon lactosylation replaces the carboxyls of the backbone, while a decrease of water concentration is also evident between 4 and 8 Å from the intraring oxygen of glucuronic acid residues and from the interglycosidic oxygens (Figure 6).

5. CONCLUSIONS

MD simulations indicate that 1-amino-1-deoxy-lactitol-grafted HA exhibits reduced flexibility of glycosidic linkages in the proximity of the grafted groups, while their conformations differ slightly from those of HA in an aqueous solution. The grafted groups favor an *extended* conformation, although the proportion of *folded* conformations increases with DS. The 1-amino-1-deoxy-lactitol-grafted groups affect the strength and residence times of hydrogen bonds, driven by the pivotal dihedral φ_{10} , whose population increases with DS, also

promoting contact between the two proximal $\beta\text{-Gal}$ moieties, and enabling a balance between intermolecular forces of the backbone and those of the grafted groups. Transient hydrogen bonds between these grafted groups through their LNK or $\beta\text{-Gal}$ subunits and the proximal glucosamine (some of which are mediated by water molecules) have been predicted, and their frequency increases slightly with DS. Finally, the size and secondary structure of grafted HAs are sensitive to the increased frequency of the hydrogen bonds cited above and, consequently, also to the crucial parameter, DS. HYLACH remains a water-soluble polymer, despite the depletion of water in the proximity of the amide groups, and in the vicinity of the intraring and interglycosidic oxygens.

In summary, multiple-independent MD simulations, combined with suitable statistical analyses, support the contention that DS is a crucial structural parameter in grafted HA polymers since this parameter modulates the size, conformation, hydrogen bond network, and secondary structure and will influence the efficiency with which carried drug moieties, such as $\beta\text{-Gal}$, will be recognized.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biomac.4c01318>.

The cartesian coordinates of oligosaccharides HA, HYLACH50, and HYLACH100 in their initial CFG A and CFG B conformations are available to download as pdb files: HA_CFGA.pdb, HA_CFGB.pdb, HYLACH50_CFGA.pdb, HYLACH50_CFGB.pdb, HYLACH100_CFGA.pdb, and HYLACH100_CFGB.pdb. The cartesian coordinates of oligosaccharides HA, HYLACH50, and HYLACH100 in their predicted conformations are available to download as pdb files: HA_PREDICTED.pdb, HYLACH50_PREDICTED.pdb, and HYLACH100_PREDICTED.pdb (PDF)

Cartesian coordinate: HA_CFGA (PDB)

Cartesian coordinate: HA_CFGB (PDB)

Cartesian coordinate: HYLACH50_CFGA (PDB)

Cartesian coordinate: HYLACH50_CFGB (PDB)

Cartesian coordinate: HYLACH100_CFGA (PDB)

Cartesian coordinate: HYLACH100_CFGB (PDB)

Cartesian coordinate: HA_PREDICTED (PDB)

Cartesian coordinate: HYLACH50_PREDICTED (PDB)

Cartesian coordinate: HYLACH100_PREDICTED (PDB)

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Funding

This research was supported by Glycore Pharma srl, Via G. Oberdan 14, 35122 Padova, and Istituto di Ricerche Chimiche e Biochimiche G. Ronzoni—Fondazione nonprofit—Via G. Colombo 81—20133 Milano.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We acknowledge Prof. R. J. Woods (University of Georgia) for the interesting suggestion concerning the parameterization of the GLYCAM06 force field. Dr. Sabrina Bertini and Dr. Ming Hong Ni of (IRCB ‘G. Ronzoni’) are thanked for their interesting and fruitful discussions. Prof. Giuseppina Raffaini (Politecnico di Milano) is thanked for the interesting and fruitful discussion.

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